EXP.NO: 5	Write a Servlet to demonstrate the difference between HTTP GET and POST methods by creating a form and handling requests accordingly.
DATE:05/03/2025	

AIM:

To demonstrate the difference between the HTTP GET and POST methods by creating a form in an HTML page and handling the form submissions using two different HTTP methods in a Java Servlet.

ALGORITHM:

1. **Input:**
    - The user fills out a web form with their name and email.
    - The form will use both GET and POST methods for data submission.

2. **Steps for Creating and Handling the HTTP GET and POST Requests:**
    1. **Create an HTML Form:**
        - Design an HTML form with two input fields: one for the name and one for the email.
        - Create two buttons to allow the user to submit the form using GET or POST methods.
        - Use the action attribute to define the Servlet URL and specify the form method (either GET or POST).
    2. **Create the Servlet to Handle GET and POST Requests:**
        - Write a Java Servlet class that will handle both GET and POST requests.
        - The servlet will have two methods:
            - doGet() to handle the GET requests.
            - doPost() to handle the POST requests.

- In both methods, retrieve the form data using request.getParameter().
- In the doGet() method, process and display the data in the URL.
- In the doPost() method, process and display the data in the HTTP request body.

3. **Handle GET Requests in the Servlet:**
    - In the doGet() method, use request.getParameter() to retrieve the name and email submitted via GET.
    - Output the data as a response, including the name and email in the URL (visible to the user).
4. **Handle POST Requests in the Servlet:**
    - In the doPost() method, use request.getParameter() to retrieve the name and email submitted via POST.
    - Output the data as a response, displaying the name and email without exposing them in the URL (more secure).
5. **Return the Output:**
    - The Servlet should return an HTML page displaying the name and email entered by the user.
    - For the GET method, the output should be visible in the URL.
    - For the POST method, the output should be visible in the response body, not the URL.
6. **Deploy and Test the Servlet:**
    - Compile the Servlet class and deploy it to a Servlet container (like Apache Tomcat).
    - Test the form using GET and POST methods to see the difference in data submission.
7. **Observe and Compare GET and POST Results:**

- Test the GET method: The form data is sent in the URL.
- Test the POST method: The form data is sent in the HTTP request body, not visible in the URL.

PROGRAM:

```
<!DOCTYPE html>
<html>
<head>
<title>Dragon Stone Restaurant</title>

<style>
body {
   margin: 0;
   font-family: "Segoe UI";
   background-image: url("image.jpg");
   background-size: cover;
   color: white;
}

header {
   background: rgba(0,0,0,0.8);
   color: #ff4d4d;
   padding: 15px;
   text-align: center;
   font-size: 24px;
}

.container {
   width: 85%;
   margin: 30px auto;
   display: flex;
   gap: 20px;
}

.card {
   flex: 1;
   background: rgba(0,0,0,0.85);
   padding: 20px;
   border-radius: 10px;
}
```

```
h3 {
   color: #ff4d4d;
}

input {
   width: 100%;
   padding: 10px;
   margin: 8px 0;
   border-radius: 5px;
   border: none;
}

button {
   width: 100%;
   padding: 10px;
   background: #ff4d4d;
   color: white;
   border: none;
   border-radius: 5px;
}
</style>
</head>

<body>

<header>🍽 Dragon Stone Restaurant</header>

<div class="container">

<!-- GET -->
<div class="card">
<h3>Search Available Tables (GET)</h3>

<form action="BookingServlet" method="get">
   <input type="date" name="date" required>
   <button type="submit">Search</button>
</form>
</div>

<!-- POST -->
<div class="card">
<h3>Book a Table (POST)</h3>

<form action="BookingServlet" method="post">
   <input type="text" name="name" placeholder="Customer Name"
required>
```

```
    <input type="tel" name="mobile" placeholder="Mobile Number" required>
    <input type="email" name="email" placeholder="Email" required>
    <input type="date" name="date" required>
    <input type="time" name="time" required>
    <input type="number" name="guests" placeholder="No. of Guests"
required>
    <button type="submit">Book Table</button>
</form>

</div>

</div>

</body>
</html>
```

Servlet5.java:

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;

@WebServlet("/BookingServlet")
public class BookingServlet extends HttpServlet {

    // GET → Search Tables
    protected void doGet(HttpServletRequest request,
                         HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        String date = request.getParameter("date");

        out.println("<html><body style='font-family:Segoe
UI;background:#111;color:white;'>");

        out.println("<div style='width:500px;margin:50px
auto;background:#222;padding:20px;border-radius:10px;text-
align:center;'>");

        out.println("<h2 style='color:#ff4d4d;'>🔍 Available Tables</h2>");
```

```
        if (date != null && !date.isEmpty()) {
            out.println("<p>Available tables on: <b>" + date + "</b></p>");
            out.println("<p>Table 1 - 2 Seats</p>");
            out.println("<p>Table 2 - 4 Seats</p>");
            out.println("<p>Table 3 - 6 Seats</p>");
        } else {
            out.println("<p>No date selected</p>");
        }

        out.println("<br><a href='index.html' style='color:red;'>Back</a>");
        out.println("</div></body></html>");
    }

    // POST → Book Table
    protected void doPost(HttpServletRequest request,
                          HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        String name = request.getParameter("name");
        String mobile = request.getParameter("mobile");
        String email = request.getParameter("email");
        String date = request.getParameter("date");
        String time = request.getParameter("time");
        String guests = request.getParameter("guests");

        out.println("<html><body style='font-family:Segoe
UI;background:#111;color:white;'>");

        out.println("<div style='width:500px;margin:50px
auto;background:#222;padding:20px;border-radius:10px;text-
align:center;'>");

        out.println("<h2 style='color:#00ff99;'>✅ Table Booked
Successfully</h2>");

        out.println("<p><b>Name:</b> " + name + "</p>");
        out.println("<p><b>Mobile:</b> " + mobile + "</p>");
        out.println("<p><b>Email:</b> " + email + "</p>");
        out.println("<p><b>Date:</b> " + date + "</p>");
        out.println("<p><b>Time:</b> " + time + "</p>");
        out.println("<p><b>Guests:</b> " + guests + "</p>");
```

```
        out.println("<br><p>Thank you for choosing Dragon Stone 🍽</p>");

        out.println("<br><a href='index.html' style='color:red;'>Back</a>");

        out.println("</div></body></html>");
    }
}
```

Output:

RESULT:

A Servlet is created to demonstrate the difference between HTTP GET and POST methods by creating a form and handling requests accordingly.